ClassAll - Original synthetic course notes

Chain rule: a short teaching chapter

This original example is fictional and may be distributed with ClassAll.

Objective A: Identify inner and outer functions in a composite expression.

If y = f(g(x)), g(x) is the inner function and f is the outer function.

For y = (3x + 1)^2, the inner expression is 3x + 1; the outer function squares it.

For y = sin(4x), the inner expression is 4x; the outer function is sine.

Objective B: Differentiate a composite function using the chain rule.

The chain rule is dy/dx = f’(g(x)) * g’(x). Multiply both derivatives.

The derivative of (2x + 1)^3 is 6(2x + 1)^2.

The derivative of (5x - 2)^2 is 10(5x - 2).

For a fresh example, the derivative of (4x + 3)^2 is 8(4x + 3).

For another example, the derivative of (3x - 1)^3 is 9(3x - 1)^2.

A frequent error is to omit the derivative of the inner expression.

Check each step and explain why both factors appear.

These notes and the prepared questions are illustrative, not validated assessments.